

Project Progress Report 4

MARI — Motion-Aware Adaptive Range Index

CSE511 • Reporting period: **Weeks 6–7**

Team: **Member A** = Shah Nadim Kamran Rian • **Member B** = Saheeb Tareque

Date: [19-Aug-2026]

1. TODO list for this period

Planned task	Owner	Target
B8. Write query, crash-recovery, and memory sections (recovery semantics + atomicity caveat)	Member B	Week 6
A8. Run query, real-SIGKILL recovery, and memory benchmarks	Member A	Week 6
B9. Write robustness & tuning sections + the statistical convention	Member B	Week 7
A9. Run tuning / heteroscedastic / hotspot / enhanced / gaps benchmarks	Member A	Week 7
B10. Positioning (incl. kinetic data structures), §2.3 application, limitations, conclusion; citation audit	Member B	Week 7
A10. Build the manuscript and package the reproducible artifact	Member A	Week 7

2. What has been done —

Member A [Shah Nadim Kamran Rian]:

- **A8 — Done.** Ran `query_bench.py`, `recovery_kill.py` (real SIGKILL), `memory_bench.py`: recovery rebuilt the table **300/300**, exact to the committed prefix, <10 ms; verification ≈ 1 /result, scan $\approx 1.2\times$; memory $1.7\times/1.2\times$.
- **A9 — Done.** Ran `tuning_bench.py`, `hetero_demo.py`, `hotspot_demo.py`, `enhanced_bench.py`, `gaps_bench.py`: adversarial drift at 500k/1.5M $\rightarrow$ **0.0051** reloc/update; heteroscedastic spread $11\times \rightarrow 4\times$; hotspot multiplicity **1938** $\rightarrow$ **13** and query over-scan $3.3\times$ lower.
- **A10 — Done.** Built the manuscript and packaged a one-command reproducible artifact (`reproduce.sh`).

Member B [Saheeb Tareque]:

- **B8 — Done.** Wrote the query-cost, crash-recovery, and memory sections, including recovery semantics (T is derived, not persisted) and the honest mid-migration atomicity caveat.
- **B9 — Done.** Wrote the robustness and tuning sections (per-region $\delta(\cdot)$ model, tuning reframe, hotspot mitigation) and the statistical convention.
- **B10 — Done.** Wrote the novelty defense (vs cracking / LSM / learned / kinetic data structures), the §2.3 application (market surveillance; SEC Limit Up–Limit Down bound as δ), limitations and conclusion; ran the citation audit (23 references).

3. Additional validation completed this period

- **Fair benchmarking** (faculty protocol: same language, same machine, CPU pinned; RSS = %M, CPU = %U+%S): peak memory MARI **37 MB** vs ART/PGM/Bx **72–74 MB** ($\sim 2\times$ less), all exact. (`run_method.py`, `fair_bench.py`, `fair_bench.sh`)
- **g–w decoupling experiment:** fixing w and raising the guard g cuts relocation **0.057** $\rightarrow$ **0.025** while scan stays flat (~ 1.22): the guard reduces relocation at *zero scan cost*. (`gw_sweep.py`)
- **New real datasets:** forex (exchange rates, 22 currencies) — 92.6% within guard, reloc 0.025, exact; oil (Brent) — a more volatile commodity. (`new_real_bench.py`)
- **Stress testing** on the most-volatile real S&P equities: MARI stays **exact** and relocates $\sim$ **3.3–8.8** $\times$ fewer than ART/PGM/Bx even under heavy tails, degrading gracefully; scan/result 1.13 vs 16–1300 for baselines. (`stress_bench.py`, `stress_compare.py`, `stress_guard.py`, `stress_scan.py`)

- **Statistical validation:** 12 seeds, 95% bootstrap CIs, paired Wilcoxon ($p < 0.01$, Holm-corrected), Cliff's $\delta = -1.0$ on relocation; all runs exact. (`stat_validate.py`)
- **Ablations:** removing the guard ($g=0$) raises relocation **2.26** $\times$ at equal scan (validates Proposition 1); ε -sweep confirms $O(1/\varepsilon)$ compaction with relocation invariant; without verification, **9.6**% of queries return false hits. (`gw_ablation.py`, `eps_verify.py`)